

Denoising Diffusion MRI with PCA and DCT

ELEC 523

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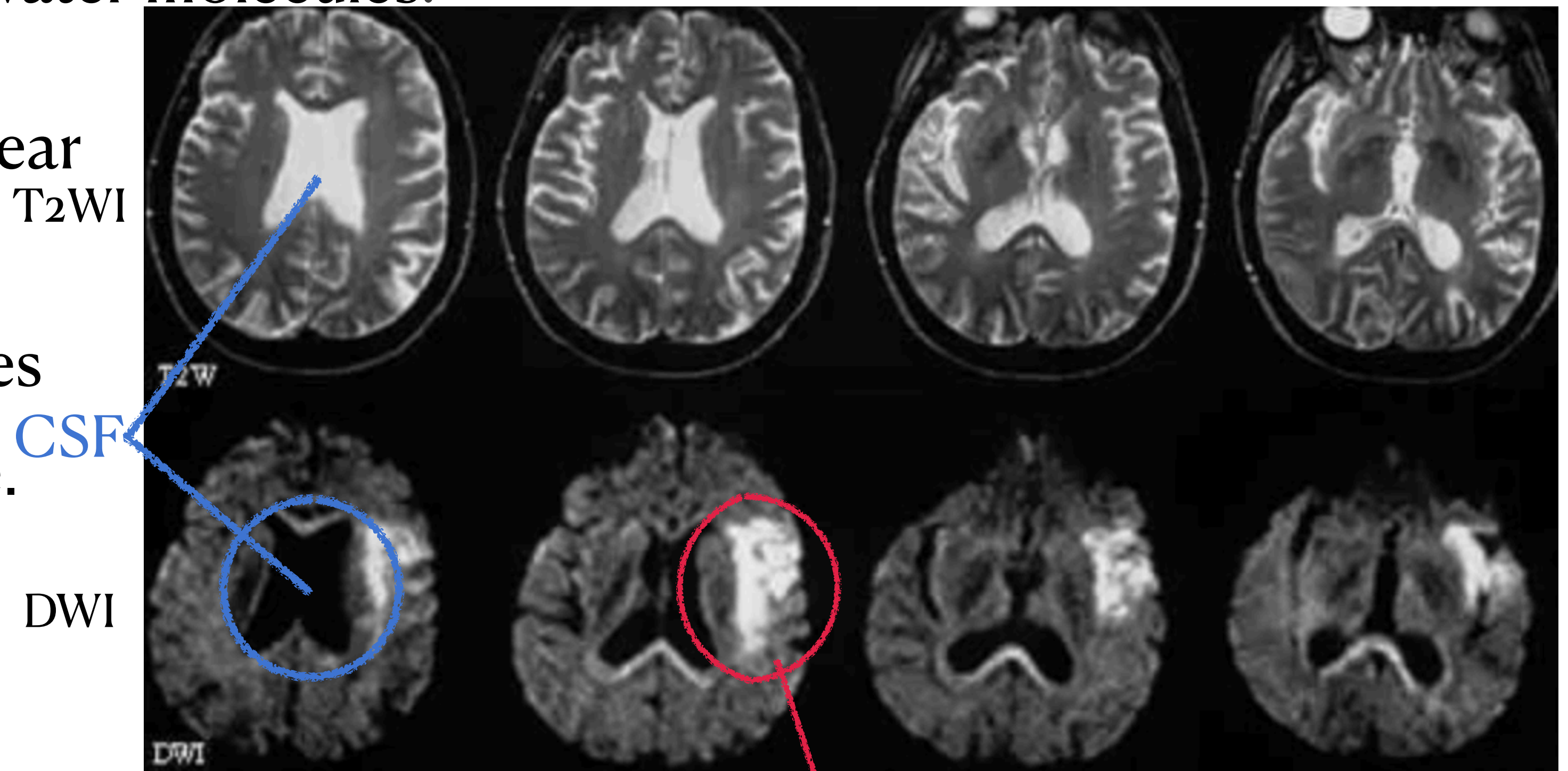
DWI

Advantage

Diffusion-weighted imaging (DWI) is a form of MR imaging based upon measuring the random Brownian motion of water molecules.

Free-moving water molecules appear low-intensity in the image.

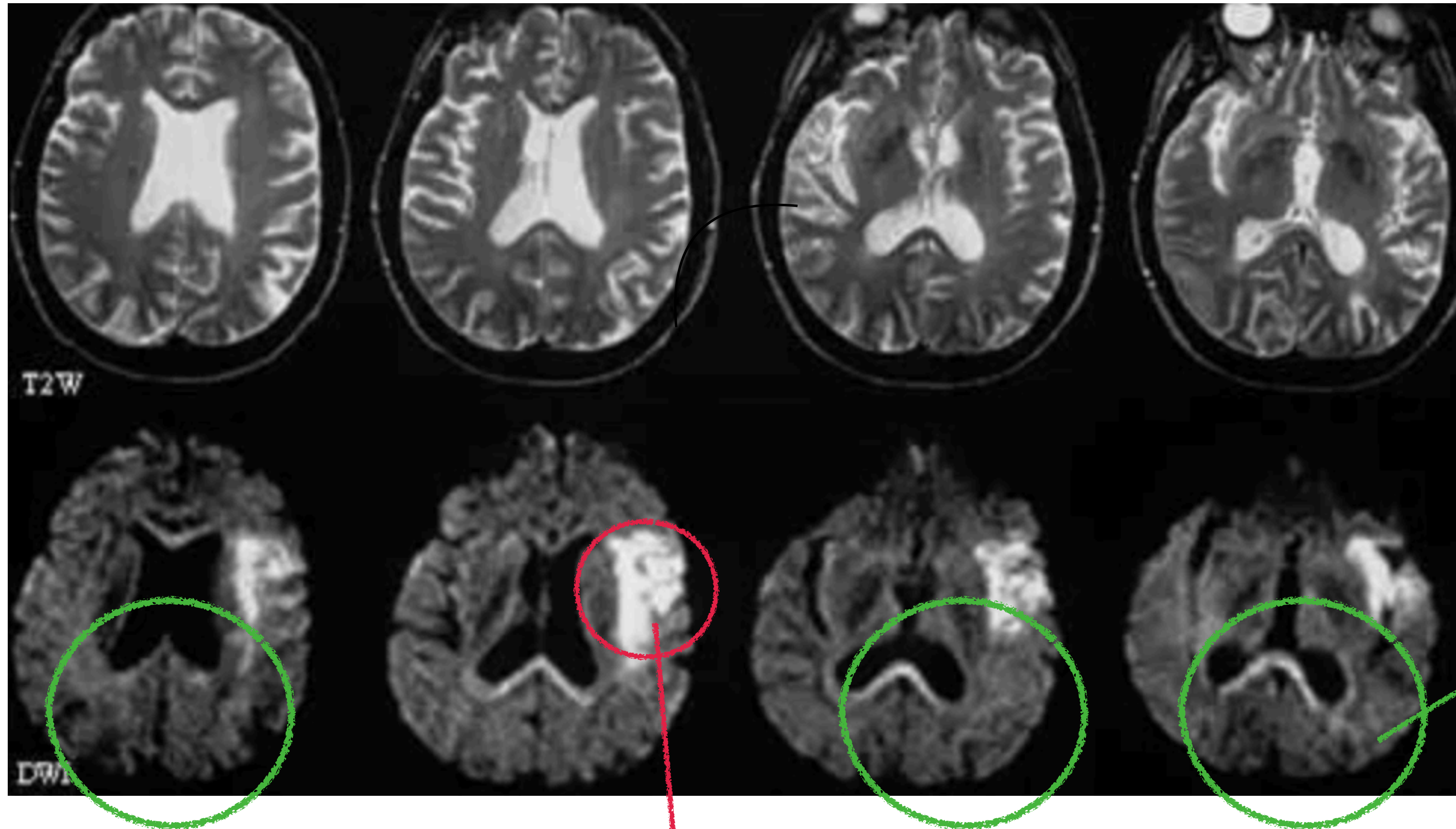
Mobility-restricted water molecules appear **high** intensity in the image.



Introduction and Motivation

Disadvantage

- T₂WI



- DWI

ischemia³

noise

Problem : Noise in DWI images caused by

1. Inhomogeneities in the magnetic field
2. Differences in magnetic susceptibility between different tissues

Goal : Reduce noise in DWI images

Method : 1. Discrete Cosine Transform (DCT) (frequency domain)
2. Principal Component Analysis (PCA) (pixel domain)

Method

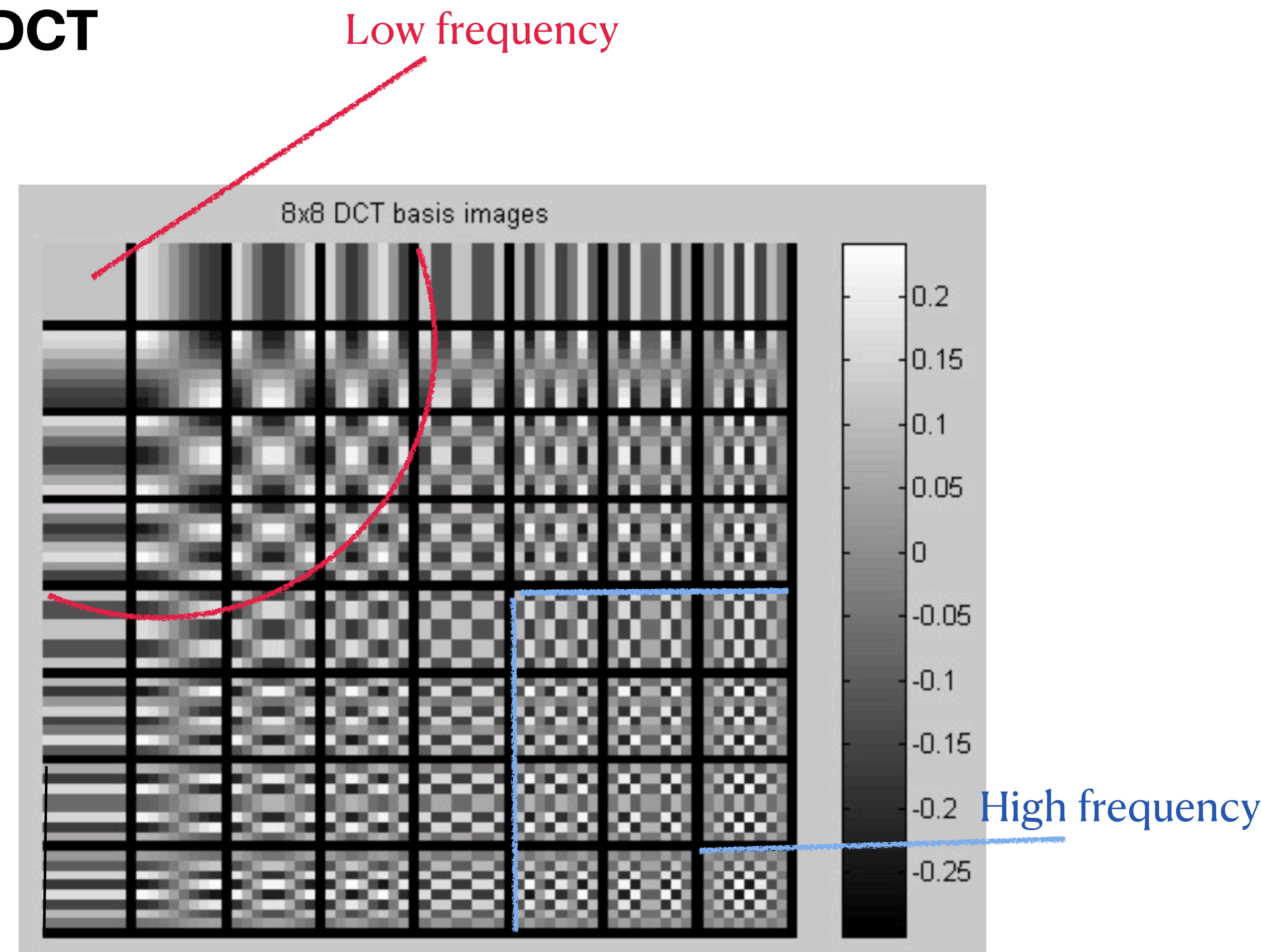
DCT

The **Discrete Cosine Transform (DCT)** is very similar to the Discrete Fourier Transform (DFT) in that it converts an image from the pixel domain to the frequency domain.

Method

DCT

- Why it works?
- The **noise** in the image is usually a **high-frequency** component, and the contour is a low-frequency component. We achieve noise reduction by removing high-frequency components and retaining low-frequency components.
- We apply a **low pass filter** to the image

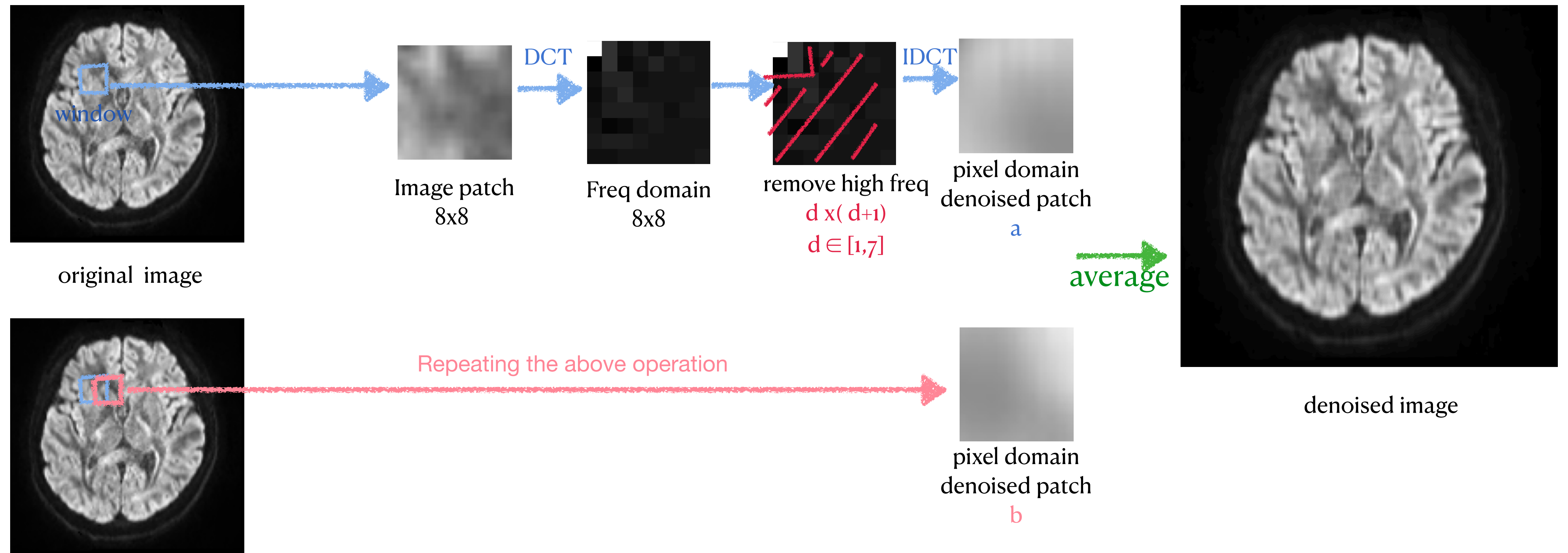


Method

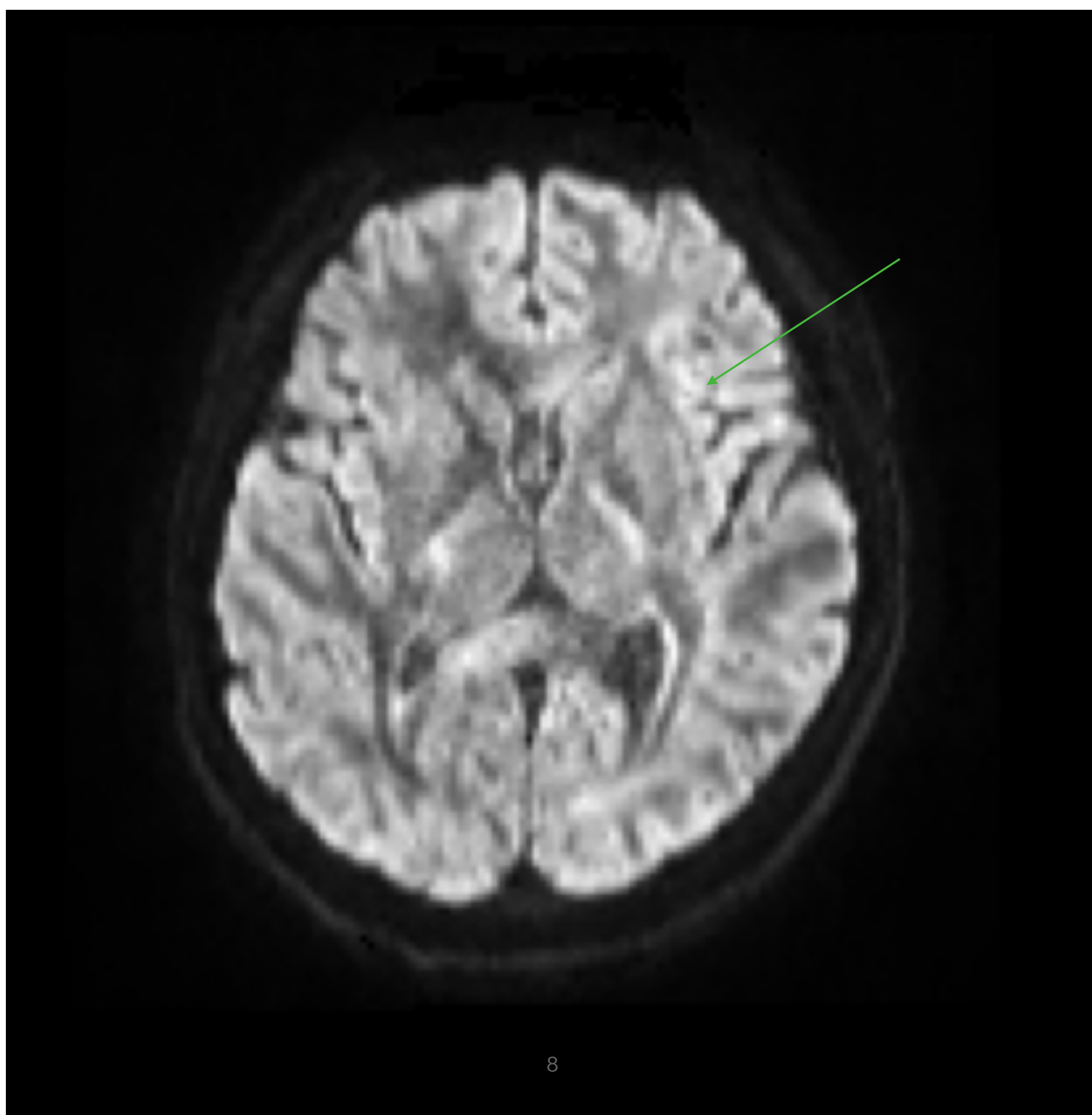
DCT flowchart

- How it works?

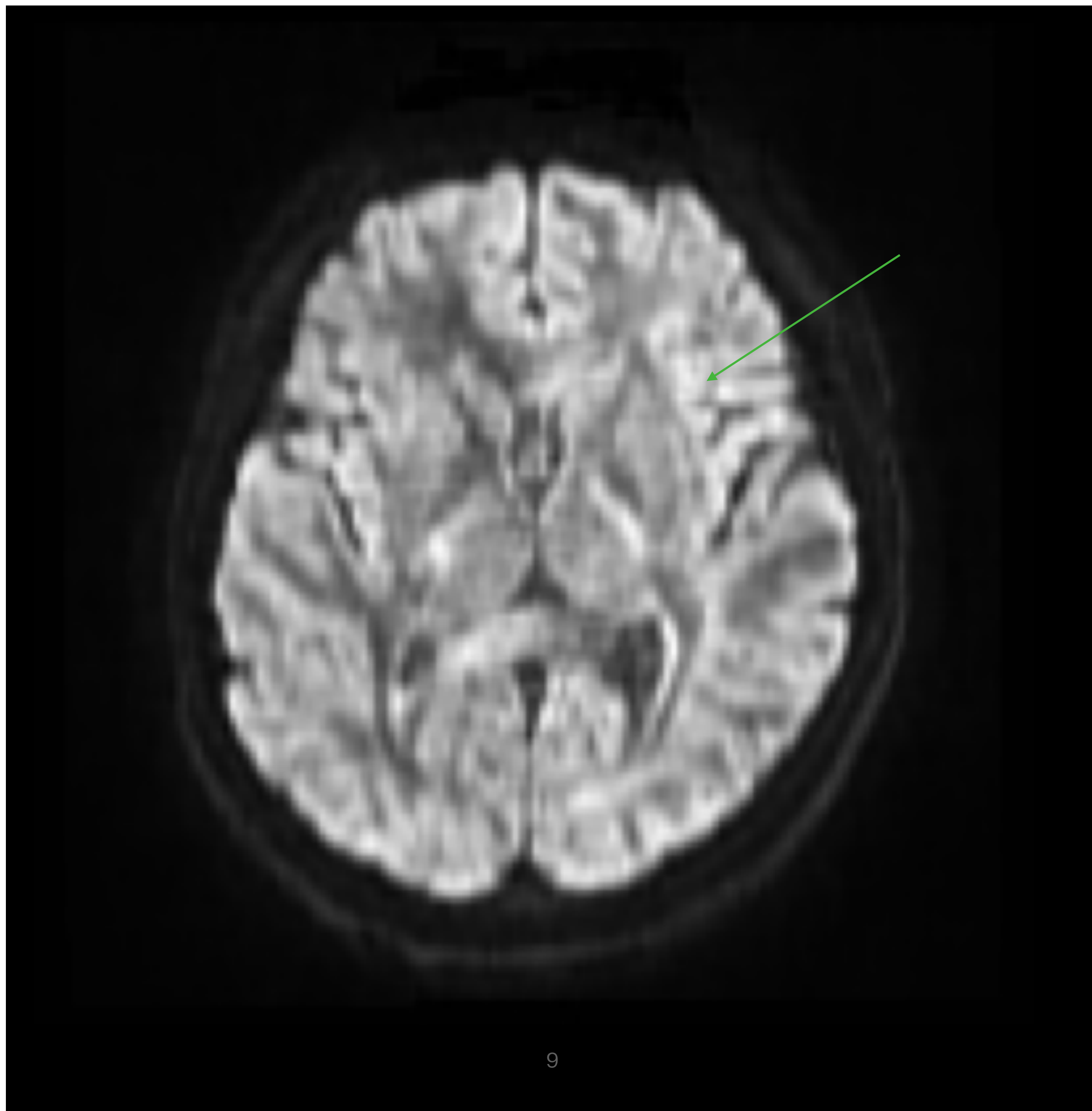
denoised by decreasing the value of d



original image



denoised image



Method

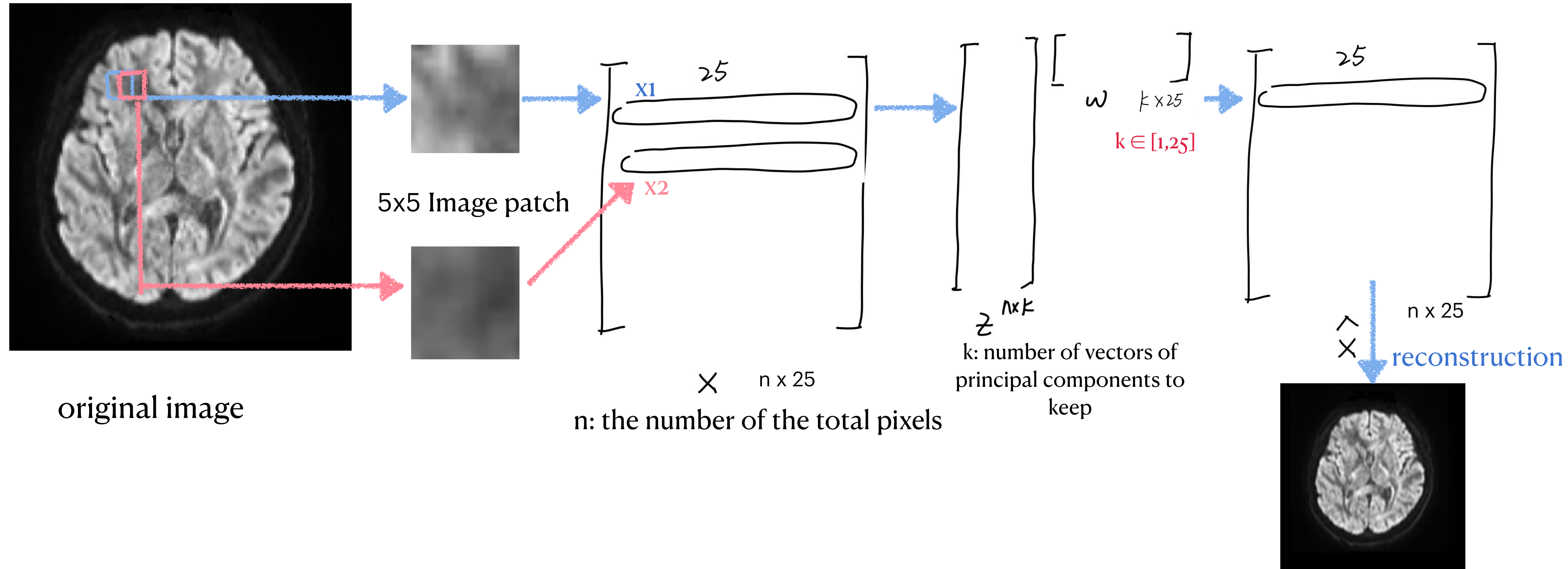
PCA

- Why it works?
- **Principal Component Analysis (PCA)** can reduce high-dimensional data to low-dimensional. This dimensionality reduction process is a “**de-correlation**” process.
- PCA tends to discard components that are less correlated with other components. **Noise** has **less correlation** with others in the image. Therefore, PCA will discard the noise components first.
- We get this idea from LPG-PCA^[1]

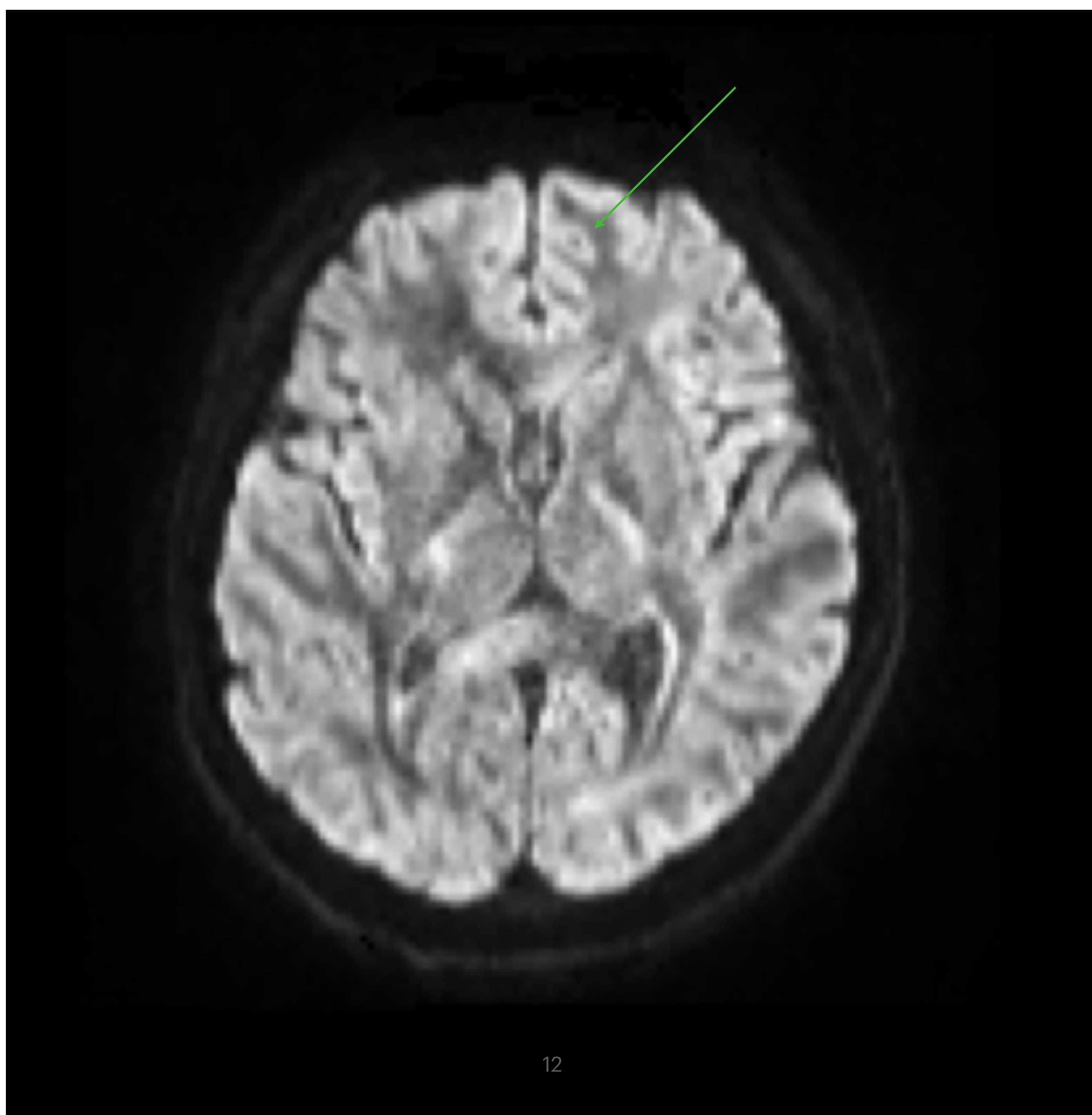
Method

- How it works?

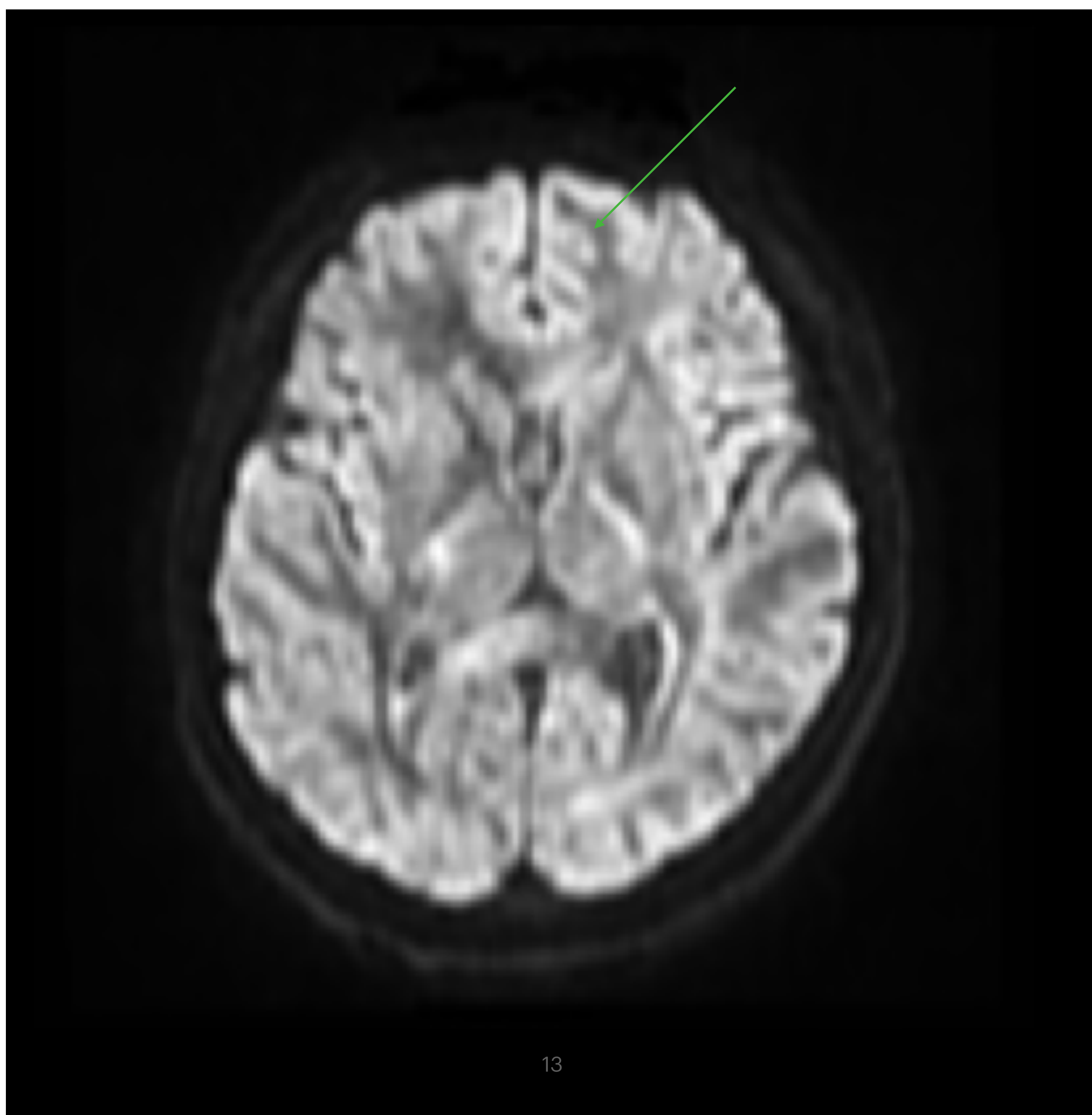
denoised by decreasing the value of k



original image



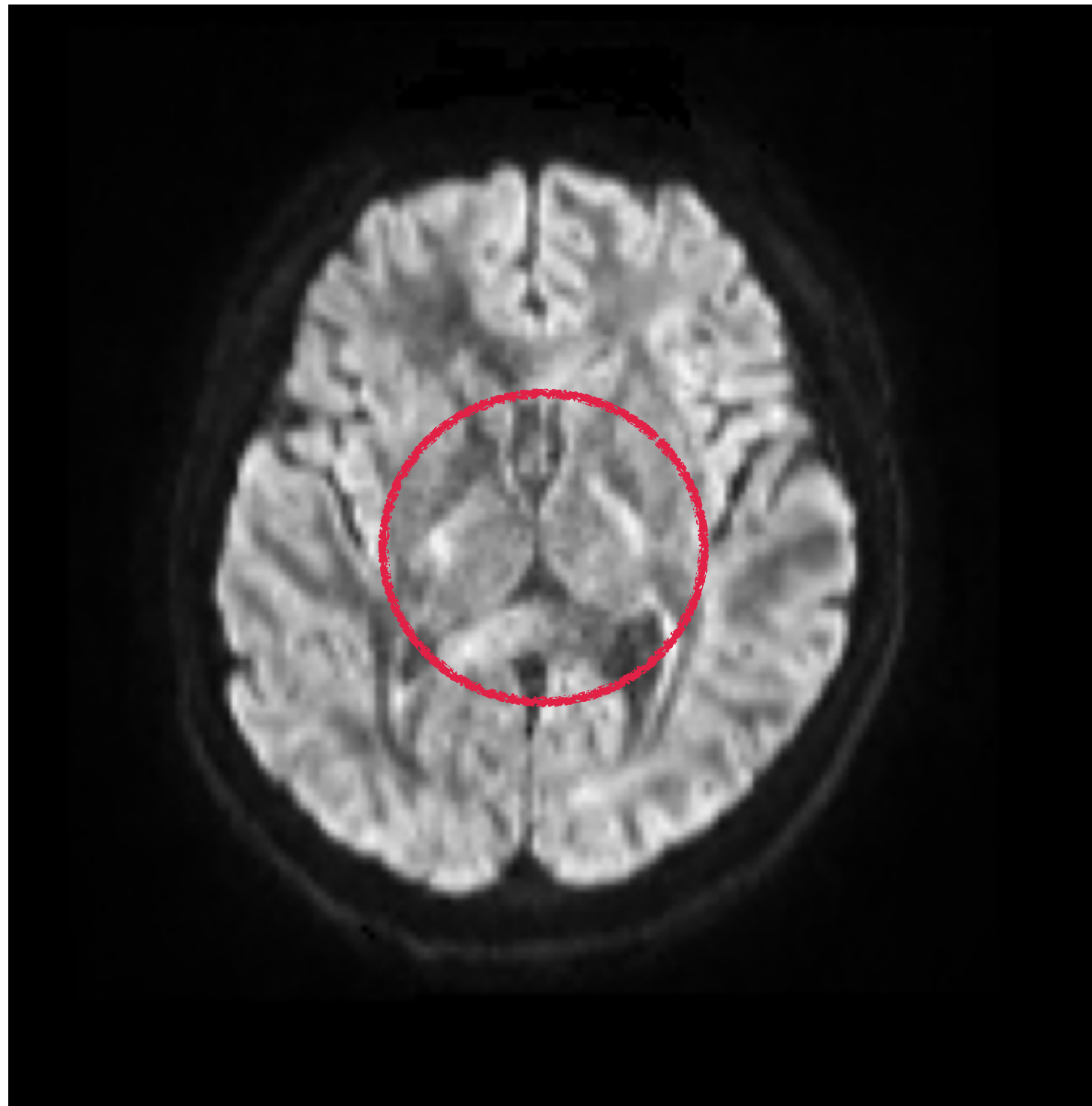
denoised image



Metric

- **Signal-to-noise ratio (SNR)** is defined as the ratio of signal power to the noise power. Higher SNR values mean less noise.
- **Structural similarity index measure (SSIM)** is used for measuring the similarity between two images. Higher SSIM values mean higher similarity. It ranges from 0 to 1, where 1 means the two images are exactly the same.

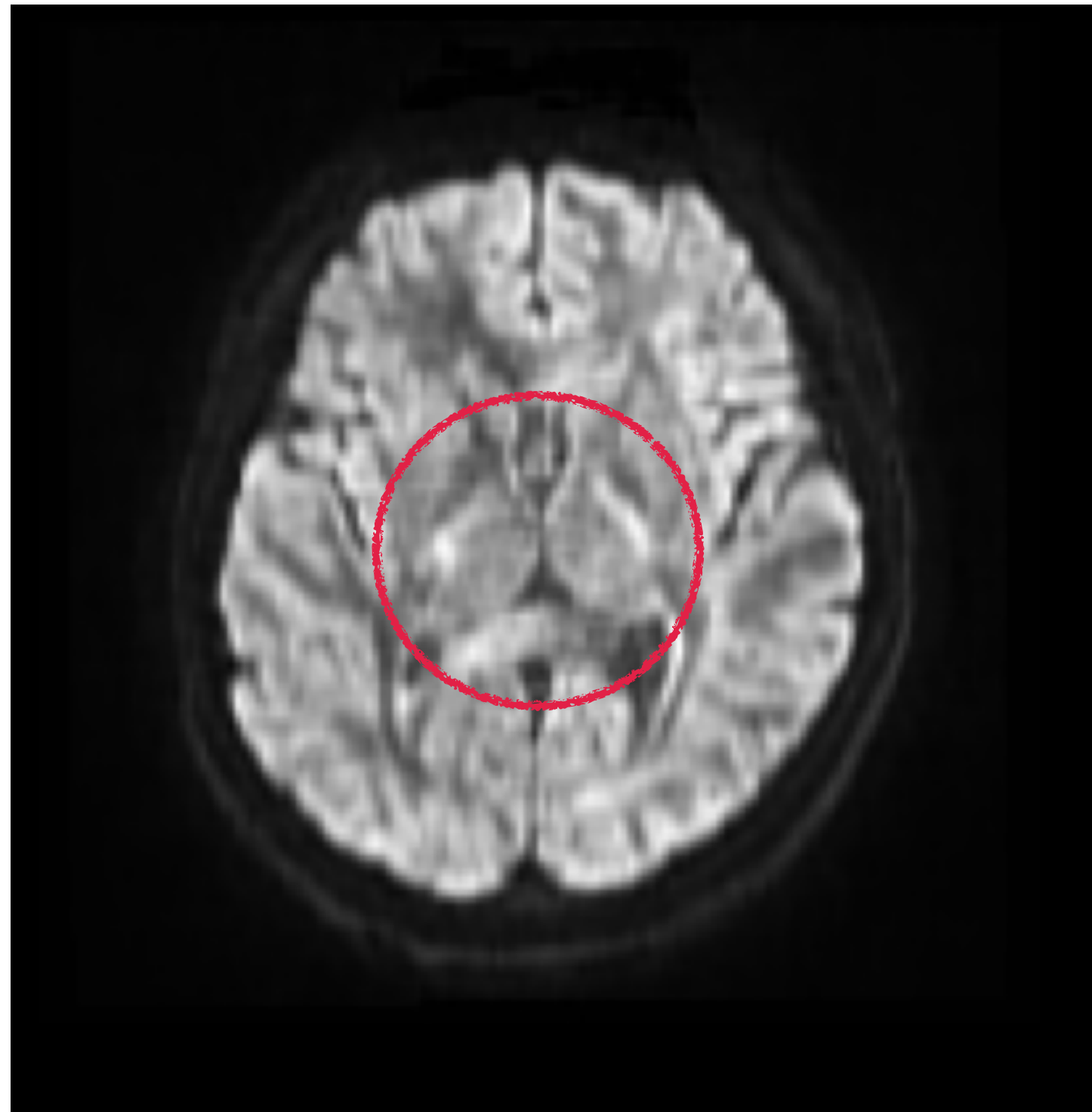
Experiment Result



original

SNR=16.5 dB

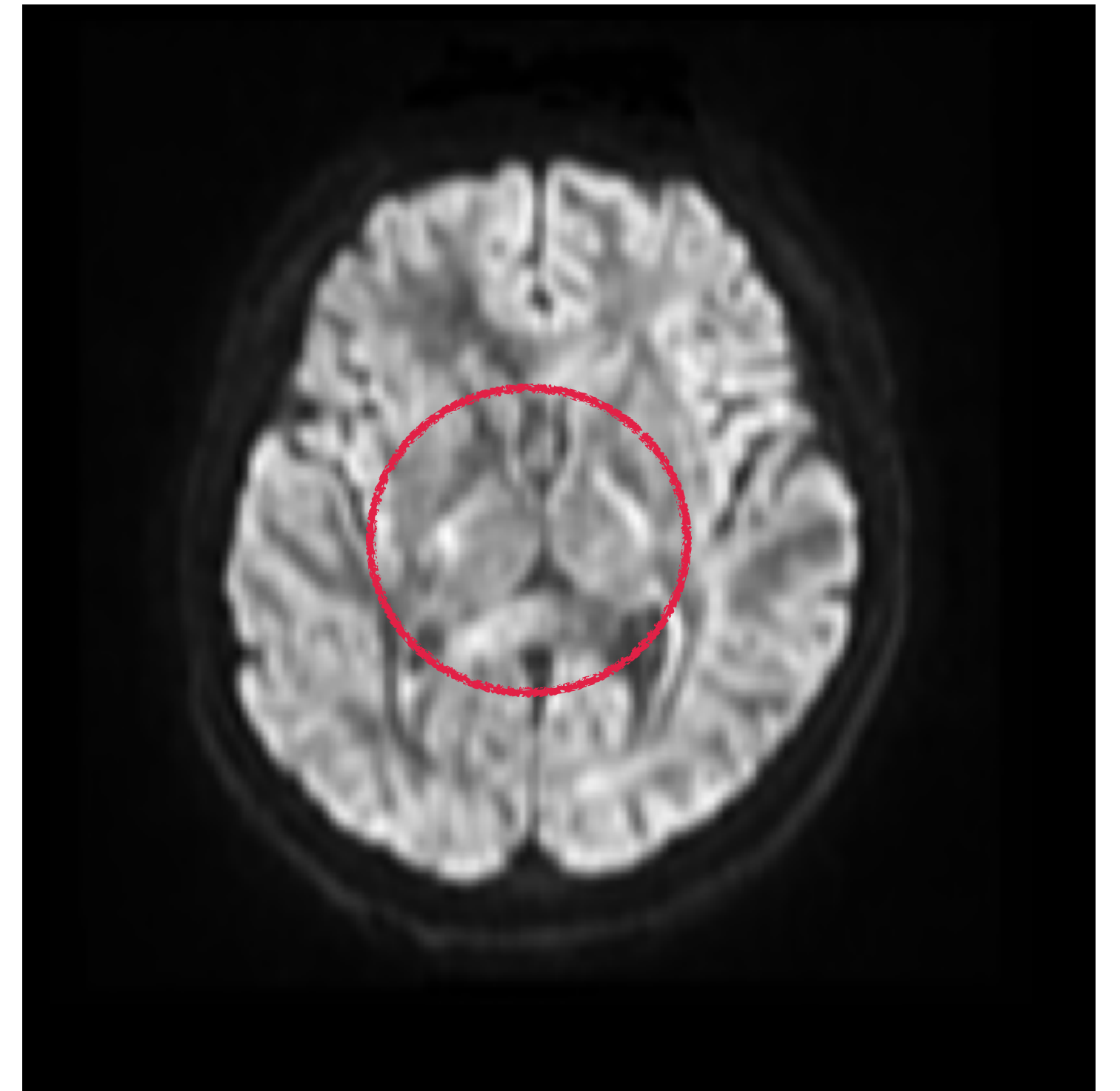
ssim = 1.0



DCT

SNR=19.1 dB

ssim = 0.90



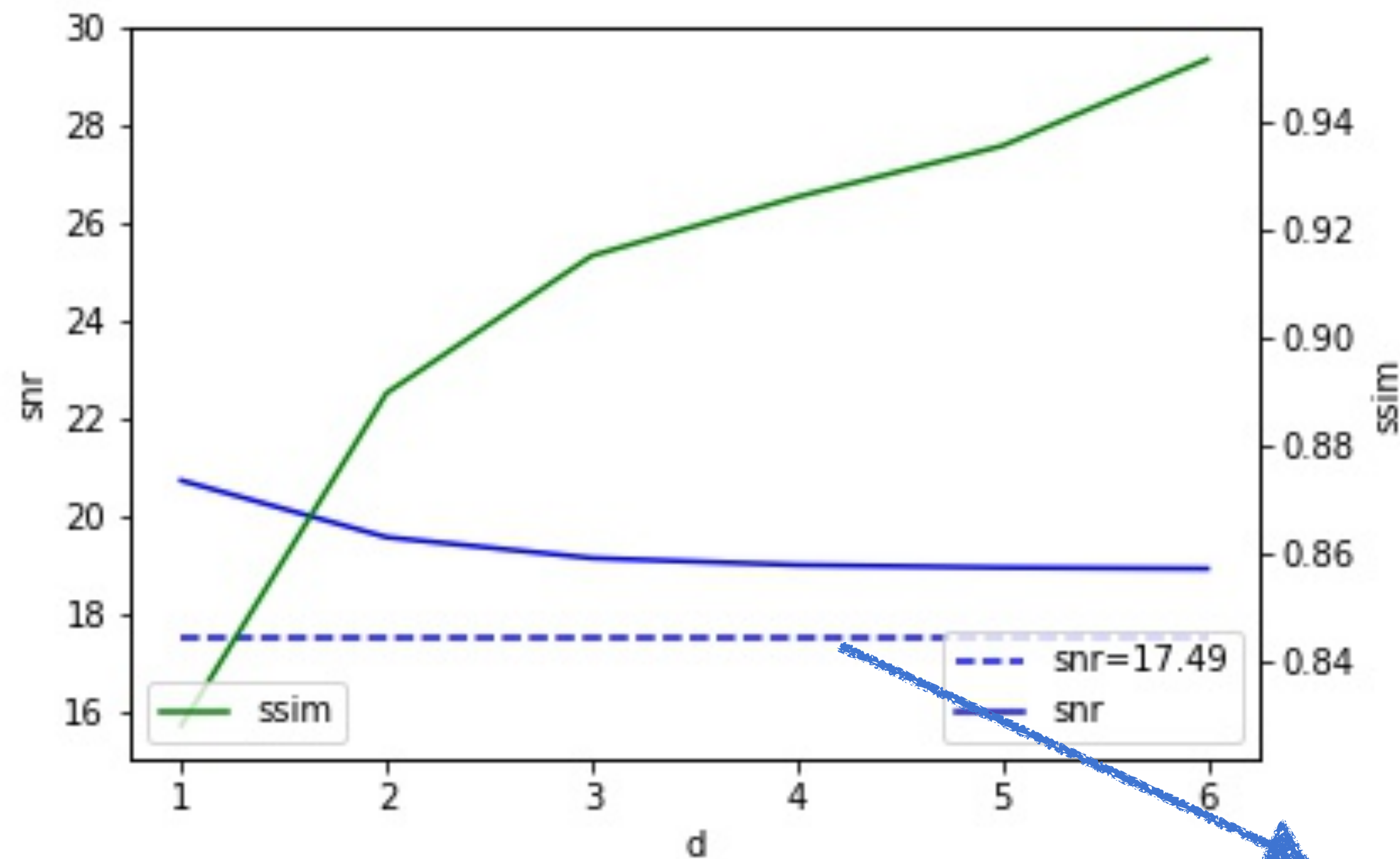
PCA

SNR=22.2 dB

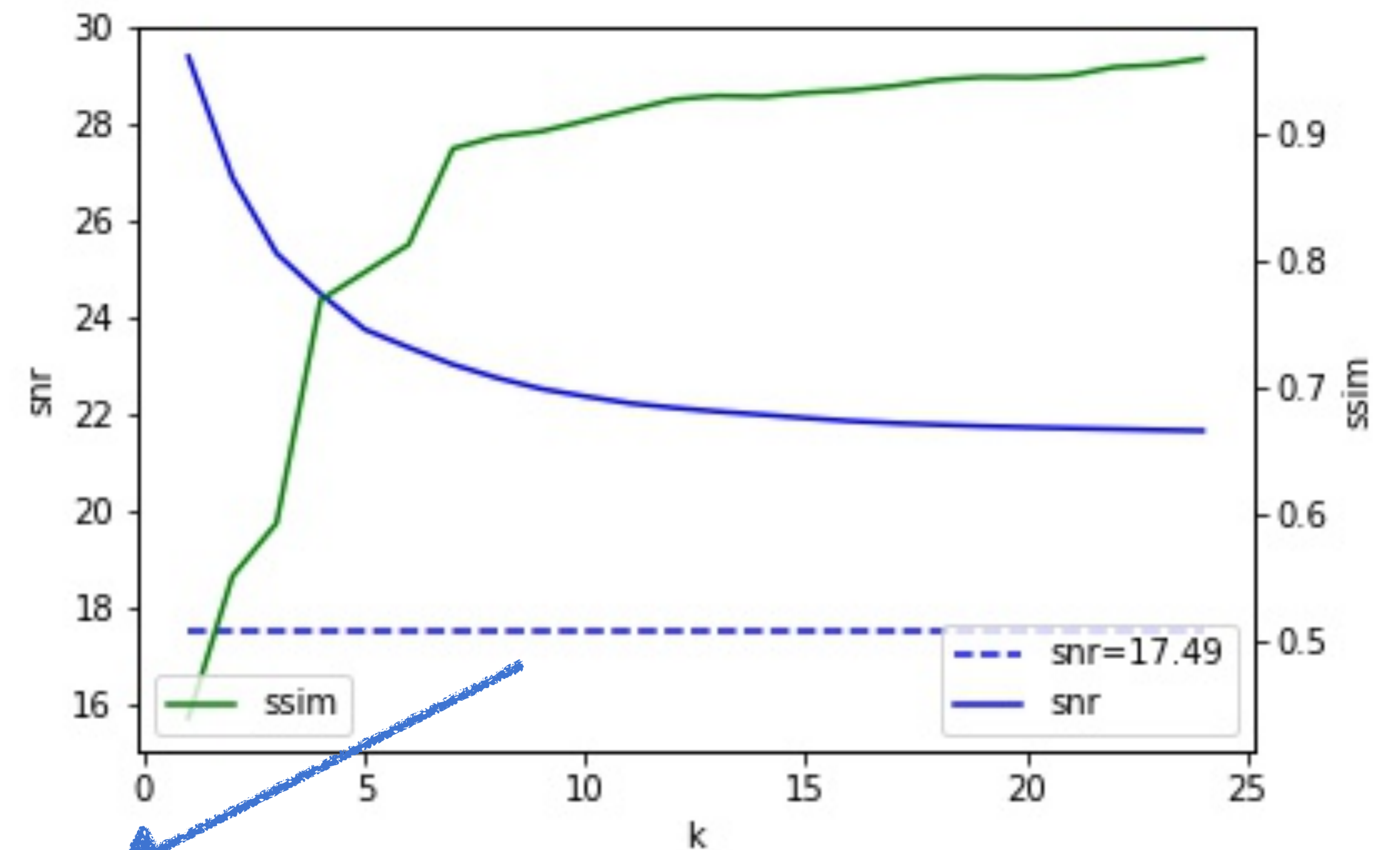
ssim = 0.90

Experiment Result

DCT



PCA



The dashed line is the SNR of the image before denoising

d: number of rows in frequency domain to keep

k: number of principle components to keep

Experiment Result

- The recommended ranges for d is [2,5] (patch size is 8x8)
- for k is [8,16] (patch size is 5x5)
- PCA can increase the SNR of the image by 4.7 ± 0.3 dB
- DCT can increase the SNR of the image by 1.7 ± 0.5 dB

	SNR(dB)	SSIM
b=0	15	1.0
b=1000	17.5	1.0
b=2000	16.5	1.0
b=3000	16.2	1.0

original

	SNR(dB)	SSIM
b=0	19.3	0.91
b=1000	22.2	0.92
b=2000	21.5	0.90
b=3000	21.2	0.92

PCA
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	SNR(dB)	SSIM
b=0	16.4	0.91
b=1000	19.1	0.91
b=2000	18.7	0.90
b=3000	18.0	0.91

DCT

The b-value reflects the strength and timing of the gradients used to generate diffusion-weighted images.

End of Presentation
Thanks!